

A Triangulation Test for Possible Thermodynamic Contributions to Gravitational Lensing in Galaxy Clusters

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ABSTRACT

We present a triangulation methodology to test whether gravitational lensing in galaxy clusters exhibits residual structure correlated with intracluster medium (ICM) thermodynamics beyond what is expected from mass alone. The method compares three independent observables: (1) projected mass from gravitational lensing, (2) dynamical mass inferred from hydrostatic equilibrium (HSE), and (3) entropy gradients derived from X-ray temperature and density profiles. We define a residual mass profile $R(r) = M_{\text{lens}}(r) - M_{\text{HSE}}(r)$ and test whether it correlates with the magnitude of the entropy gradient $|\nabla K(r)|$.

This approach provides a direct observational test of phenomenological models in which lensing convergence may include a small additional term coupled to thermodynamic structure. We outline order-of-magnitude consistency requirements that any phenomenological coupling of this type would need to satisfy in order to remain compatible with existing precision tests of gravity and cosmology. We apply the triangulation method to two pilot systems: the merging Bullet Cluster and the relaxed control cluster Abell 1835. No statistically significant correlation is detected in either system (Bullet: $\rho = -0.129$, $p = 0.21$), consistent with the expected sensitivity limits of current data for per cent-level effects.

The methodology is fully general and can be applied to larger cluster samples and to future high-precision lensing and X-ray data sets. As a speculative extension, we outline a possible line-of-sight redshift signal associated with thermodynamic gradients that could serve as an independent falsification criterion for this class of phenomenological couplings.

Key words: galaxies: clusters: general – gravitational lensing: weak – gravitational lensing: strong – X-rays: galaxies: clusters – cosmology: observations – methods: data analysis

1 INTRODUCTION

Galaxy clusters provide multiple independent probes of their mass distribution. Gravitational lensing measures the total projected mass through its effect on photon trajectories, while X-ray observations of the intracluster medium (ICM) allow dynamical mass estimates under the assumption of hydrostatic equilibrium (HSE). In relaxed systems these mass estimates are broadly consistent, but in merging clusters significant discrepancies are frequently observed (Zhang et al. 2010; Mahdavi et al. 2013; Applegate et al. 2014; von der Linden et al. 2014; Hoekstra et al. 2015).

These differences are generally attributed to non-equilibrium processes, non-thermal pressure support, and projection effects (Nagai, Kravtsov & Vikhlinin 2007; Lau, Kravtsov & Nagai 2009; Nelson, Lau & Nagai 2014; Biffi et al. 2016; Eckert et al. 2019).

At the same time, galaxy clusters exhibit large spatial variations in thermodynamic structure. Shock fronts, cold fronts, and mixing layers create strong entropy gradients that trace irreversible energy redistribution in the ICM. These gradients are especially prominent in merging systems, where deviations from equilibrium are largest.

In this work we introduce an observational methodology designed to test whether residual differences between lensing mass and HSE mass estimates trace the thermodynamic structure of the ICM. Specif-

ically, we define the mass residual

$$R(r) = M_{\text{lens}}(r) - M_{\text{HSE}}(r) \quad (1)$$

and examine whether it correlates with the magnitude of the entropy gradient

$$|\nabla K(r)|, \quad K = T n_e^{-2/3}. \quad (2)$$

Such a correlation would not be expected in standard analyses where lensing responds only to the mass distribution and HSE deviations are driven purely by dynamical state. However, several phenomenological approaches inspired by thermodynamic analogies to gravity (Jacobson 1995; Padmanabhan 2010; Verlinde 2011) have suggested that gravitational response could, in principle, include small contributions linked to entropy structure.

Rather than adopting any specific theoretical model, we treat this possibility as a parameterized observational ansatz and construct a test that is agnostic to the underlying mechanism.

The goal of this paper is therefore methodological: to establish a reproducible triangulation test that can be applied to current and future data sets to search for, or constrain, any additional coupling between lensing convergence and ICM thermodynamics. We encourage interpreting the analysis as a null test of standard assumptions, where detection of a statistically significant correlation would motivate further theoretical investigation, and non-detection would constrain this class of phenomenological extensions.

We first outline the formalism of the triangulation method, then

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present consistency requirements ensuring that any coupling explored remains compatible with existing precision constraints. Finally, we apply the method to two well-studied systems to demonstrate feasibility and sensitivity.

2 TRIANGULATION METHODOLOGY

2.1 Three Independent Observables

The triangulation test relies on combining three measurements in each cluster:

2.1.1 Observable 1: Lensing Mass (Geometric)

Data: Weak lensing (background galaxy shear γ) and strong lensing (multiple images).

Method: Weak lensing measures ellipticities of background galaxies; the shear field is inverted to reconstruct convergence $\kappa(\theta)$. Strong lensing constrains the mass distribution in central regions via multiply-imaged background sources. Combined analyses use parametric fits (e.g. NFW profiles) or non-parametric reconstructions (Merten et al. 2009).

Enclosed mass:

$$M_{\text{lens}}(< r) = \pi r^2 \Sigma_{\text{crit}} \bar{\kappa}(< r) \quad (3)$$

where $\Sigma_{\text{crit}} = (c^2/4\pi G) \times (D_s/D_l D_{ls})$ is the critical surface density.

Data sources: JWST NIRCam, HST ACS/WFC3, Subaru Hyper Suprime-Cam, DECam.

Systematic uncertainties: Shape measurement bias ($\sim 1\text{--}3$ per cent), photometric redshift errors (~ 5 per cent), intrinsic alignments (~ 2 per cent), mass-sheet degeneracy (broken by combining strong and weak lensing). Total: $\sim 5\text{--}10$ per cent on M_{lens} (Umetsu 2020).

2.1.2 Observable 2: Hydrostatic Mass (Dynamic)

Data: X-ray surface brightness $I_X(r)$ and temperature $T(r)$ from Chandra or XMM-Newton.

Method: Assume spherical hydrostatic equilibrium:

$$\frac{dP}{dr} = -\rho_{\text{gas}} \frac{G M(< r)}{r^2} \quad (4)$$

For ideal gas with pressure $P = n_e kT$, solve for enclosed mass:

$$M_{\text{HSE}}(< r) = -\frac{kT r}{\mu m_p G} \left(\frac{d \ln n_e}{d \ln r} + \frac{d \ln T}{d \ln r} \right) \quad (5)$$

where $\mu \approx 0.6$ is the mean molecular weight for solar-metallicity ICM.

Data sources: Chandra ACIS, XMM-Newton EPIC, ACCEPT catalog (Cavagnolo et al. 2009): 239 clusters with radial T , n_e profiles.

Systematic uncertainties: Temperature deprojection (~ 5 per cent), absolute temperature calibration (~ 5 per cent), non-thermal pressure support (10–30 per cent depending on dynamical state), clumping bias (~ 5 per cent; Eckert et al. 2015). Total: $\sim 10\text{--}20$ per cent on M_{HSE} (Ettori et al. 2019).

Key assumption: Equation (4) assumes static equilibrium. In merging clusters, bulk motions and shocks produce non-thermal pressure that HSE mass estimates do not account for. This is the standard explanation for $M_{\text{lens}} > M_{\text{HSE}}$.

2.1.3 Observable 3: Entropy Gradient (Thermodynamic)

Data: Same X-ray observations as Observable 2.

Definition: ICM entropy (per particle):

$$K(r) \equiv T(r) n_e(r)^{-2/3} \quad (6)$$

This is proportional to specific entropy $s \sim k \ln(T^{3/2}/n_e)$ in an ideal gas.

Entropy gradient:

$$|\nabla K(r)| \approx \left| \frac{dK}{dr} \right| \quad (7)$$

computed from radial profiles, or $|\nabla K(x, y)|$ from 2D thermodynamic maps (Million & Allen 2008).

Physical interpretation: Entropy is approximately conserved in adiabatic flow. Shocks produce entropy jumps (irreversible heating): $K_2/K_1 = (T_2/T_1)/(n_2/n_1)^{2/3}$. Mixing and turbulence increase entropy. Entropy gradients mark sites of irreversible thermodynamic processes.

Systematic uncertainties: Propagated from T and n_e errors, deprojection effects (especially in 2D maps). Total: $\sim 20\text{--}30$ per cent on $|\nabla K|$ (Pratt et al. 2019).

2.2 The Triangulation Relation

Core test: Compute the mass residual:

$$R(r) \equiv M_{\text{lens}}(r) - M_{\text{HSE}}(r) \quad (8)$$

and test for correlation with entropy gradient:

$$\rho(R, |\nabla K|) = ? \quad (9)$$

Expected behaviour:

- **Λ CDM (null hypothesis):** $\rho \approx 0$ or weak. R arises from non-thermal pressure (bulk motions, turbulence). May coincide with shocks (which also increase entropy), but no systematic correlation with spatial structure of $|\nabla K|$.

- **Thermodynamic coupling:** $\rho > 0$, significant. If $\kappa \propto \nabla^2 K$, then M_{lens} is enhanced in regions of high $|\nabla K|$. HSE is unaffected, so $R \propto |\nabla K|$.

Why this works:

- (i) M_{lens} measures total lensing (including any thermodynamic contribution if it exists).

- (ii) M_{HSE} measures the gravitational mass inferred from force balance under the assumption that only standard gravitational forces enter the equilibrium equation.

- (iii) Residual R captures differences between lensing-inferred mass and hydrostatic equilibrium mass estimates, which may arise from non-thermal pressure support, projection effects, departures from equilibrium, or other systematic effects. Any additional contribution from thermodynamic-gravitational coupling, if present, would appear on top of these standard sources of mismatch.

2.3 Implementation

Step 1: Acquire data

- Lensing: $\kappa(\theta)$ map or parametric NFW fit (M_{200} , c_{200})
- X-ray: $T(r)$, $n_e(r)$ radial profiles (or 2D maps if available)

Step 2: Compute masses

- $M_{\text{lens}}(< r)$ from Equation (3)

- $M_{\text{HSE}}(< r)$ from Equation (5)

Step 3: Compute entropy gradient

- $K(r)$ from Equation (6)
- $|\nabla K(r)|$ from Equation (7)

Step 4: Correlation analysis

For 1D radial analysis:

- Normalize: $R_{\text{norm}}(r) = R(r)/M_{\text{lens}}(r)$
- Normalize: $G_{\text{norm}}(r) = |\nabla K(r)|/\max |\nabla K|$
- Compute: Pearson correlation coefficient $\rho_P(R_{\text{norm}}, G_{\text{norm}})$
- Compute: Spearman rank correlation ρ_S (robust to outliers)
- Test significance: p -value under null hypothesis $H_0 : \rho = 0$

For 2D spatial analysis (when available):

- Pixel-by-pixel correlation between $\kappa(x, y)$ residual and $|\nabla K(x, y)|$
- Cross-correlation function
- Spatial filtering to isolate scales of interest (~ 100 kpc)

Step 5: Control comparison

- Apply same analysis to relaxed cluster (e.g. Abell 1835, A2029)
- Compare: $|\rho_{\text{merger}}|$ vs $|\rho_{\text{relaxed}}|$
- Thermodynamic coupling hypothesis predicts: stronger correlation in mergers (high $|\nabla K|$, strong non-equilibrium) than relaxed (low $|\nabla K|$, near-equilibrium)

3 ORDER-OF-MAGNITUDE CONSISTENCY REQUIREMENTS

Any phenomenological coupling between lensing convergence and ICM entropy must remain consistent with existing high-precision tests of gravity and cosmology. We perform order-of-magnitude checks to ensure that a coupling of the form $\kappa \sim \kappa_{\text{GR}} + \alpha \nabla^2 K$, where α is treated as an effective phenomenological coefficient, is not already ruled out.

In this parameterization α implicitly carries the units required to map entropy curvature ($\text{keV cm}^2 \text{kpc}^{-2}$) into an effective surface mass density contribution; we do not assign it a fundamental origin here. The illustrative value $\alpha \sim 0.01$ is motivated by the dimensional estimate $\kappa_{\text{thermo}}/\kappa_{\text{GR}} \sim (v/c)^2 \sim 10^{-2}$ for ICM bulk velocities $v \sim 1000 \text{ km s}^{-1}$.

Important caveat: These are illustrative consistency checks, not rigorous derivations. A complete covariant theory would need to specify Lagrangian or field equations and derive predictions in all relevant regimes. We have not done this. What follows is an assessment of whether $\alpha \sim 0.01$ is obviously inconsistent with existing data.

3.1 Achromaticity of Lensing

Requirement: Lensing deflection angle α and convergence κ must be independent of photon wavelength.

Data: H0LiCOW sample: 6 lensed quasars with time delays and multi-band imaging (Suyu et al. 2017; Wong et al. 2020).

Test statistic:

$$\chi^2_{\text{achrom}}/\text{dof} = \sum_i [(\kappa_{\text{optical}} - \kappa_{\text{NIR}})^2 / \sigma^2] / N_{\text{systems}} \quad (10)$$

Result: $\chi^2/\text{dof} = 0.91$ ($N = 6$).

Interpretation: No evidence for wavelength-dependent lensing at ~ 10 per cent precision.

Consistency check: If coupling is to a scalar entropy field (not photon energy), achromaticity is expected. ✓ Consistent.

3.2 CMB Spectral Distortions

Requirement: Early-universe thermodynamics ($z \sim 10^3$) must produce standard blackbody spectrum.

Data: COBE/FIRAS (Fixsen et al. 1996; Mather et al. 1999): $|\Delta y| < 1.5 \times 10^{-5}$, $|\Delta \mu| < 9 \times 10^{-5}$.

Interpretation: CMB thermalization unaffected at sub-per cent level.

Consistency check: If thermodynamic-gravitational coupling becomes negligible at high symmetry / low entropy contrast ($z > 1000$), this is satisfied by construction. However, we have not demonstrated this from field equations. A full theory must explicitly show vanishing coupling in the radiation-dominated era. Δ Assumed consistent, not proven.

3.3 Weak Equivalence Principle

Requirement: Test masses of different compositions fall at the same rate.

Data: Lunar laser ranging (Williams, Turyshev & Boggs 2004), MICROSCOPE satellite (Touboul et al. 2017): $\epsilon < 10^{-13}$ for Earth–Moon system, $\epsilon < 10^{-15}$ for test masses in orbit.

Interpretation: No detectable composition-dependent acceleration.

Consistency check: If coupling is to total energy-momentum tensor $T^{\mu\nu}$ (not baryon number or other conserved charges), WEP is preserved. This is a structural assumption, not a derivation. Δ Assumed consistent by construction.

3.4 Big Bang Nucleosynthesis

Requirement: Expansion rate at $z \sim 10^9$ must match standard Friedmann cosmology to preserve BBN light element abundances.

Data: Planck CMB acoustic peaks + primordial deuterium abundance (Planck Collaboration 2020; Cooke, Pettini & Steidel 2018): $N_{\text{eff}} = 2.99 \pm 0.17$ (consistent with 3.046 from Standard Model).

Interpretation: No additional relativistic degrees of freedom, expansion rate normal.

Consistency check: If coupling is negligible in the radiation-dominated era (entropy gradients small, universe highly symmetric), BBN proceeds as standard. Δ Not derived; assumed by regime of applicability.

3.5 Gravitational Time Delays

Requirement: Gravitational time delay $\tau = (1 + z_d) \Delta t_{\text{geom}} \times [(1 + \phi/c^2)]$ must be independent of photon energy.

Data: H0LiCOW + TDCOSMO (Millon et al. 2020): $\Delta\chi^2/\text{dof} = 8.7 \times 10^{-4}$ ($N = 6$ systems $\times$ multiple bands).

Interpretation: Time delays are achromatic at sub-per cent precision.

Consistency check: If lensing potential Ψ (which determines time delay) is achromatic, this follows. Coupling $\kappa \propto \nabla^2 K$ affects Ψ but not in a wavelength-dependent way. ✓ Consistent.

Table 1. Summary of consistency requirements. ✓ indicates demonstrable compatibility with data; Δ indicates plausible compatibility based on regime of applicability, but not derived from field equations.

Test	Observable	Data	Status
Achromaticity	$\kappa(\lambda)$	H0LiCOW	✓
CMB spectrum	$\Delta y, \Delta\mu$	FIRAS	Δ
WEP	ϵ	LLR+MICROSCOPE	Δ
BBN	N_{eff}	Planck+D/H	Δ
Time delays	$\tau(\lambda)$	TDCOSMO	✓
Void lensing	κ_{void}	DES Y3	Δ

3.6 Void Lensing (Sign Test)

Requirement: Low-density regions (cosmic voids) must show lensing signal consistent with underdensity.

Data: DES Y3 void catalog (Fang et al. 2019; Hamaus et al. 2020): $\kappa_{\text{void}}/\kappa_{\Lambda\text{CDM}} = 0.79 \pm 0.15$.

Interpretation: Voids exhibit underdensity lensing signal, though somewhat weaker than simulations predict.

Consistency check: If voids have lower entropy than their surroundings ($\nabla^2 K < 0$), coupling would predict $\kappa_{\text{thermo}} < 0 \rightarrow$ weaker total lensing. Observed signal is qualitatively in this direction. Δ Intriguing but not conclusive (statistical significance marginal; void profiles complex).

3.7 Summary

Interpretation: A phenomenological coupling $\kappa \propto \nabla^2 K$ with $\alpha \sim 0.01$ is not obviously ruled out by these constraints. However, we emphasize this is an order-of-magnitude consistency check, not a rigorous derivation. A complete covariant formulation would be required to derive predictions self-consistently across all regimes. The present work does not attempt such a derivation and instead treats the coupling purely as an observational parameterization.

4 PILOT APPLICATION

To demonstrate the feasibility and current sensitivity of the triangulation method, we apply it to two well-studied systems: the merging Bullet Cluster (1E 0657–56) and the relaxed cool-core cluster Abell 1835.

4.1 Data Sources

For each system we combine:

- **Lensing mass profiles** derived from published weak and strong lensing reconstructions
- **Hydrostatic mass profiles** inferred from X-ray temperature and density measurements under the assumption of hydrostatic equilibrium
- **Entropy profiles** constructed from the same X-ray data via $K(r) = T(r) n_e(r)^{-2/3}$

All profiles are interpolated on to common radial bins before computing residuals and gradients.

4.2 Residual and Gradient Construction

We define the fractional mass residual

$$R_{\text{norm}}(r) = \frac{M_{\text{lens}}(r) - M_{\text{HSE}}(r)}{M_{\text{lens}}(r)} \quad (11)$$

and the normalized entropy-gradient amplitude

$$G_{\text{norm}}(r) = \frac{|\nabla K(r)|}{\max(|\nabla K|)}. \quad (12)$$

Correlations between R_{norm} and G_{norm} are quantified using both Pearson and Spearman statistics.

4.3 Bullet Cluster (Merging System)

The Bullet Cluster exhibits strong shock structures and pronounced entropy gradients associated with its merger geometry. Lensing data are from JWST NIRCcam (Cha et al. 2025; Cho et al. 2025); X-ray data are from Chandra 500 ks (Million & Allen 2008) and ACCEPT profiles (Cavagnolo et al. 2009).

We find:

- Pearson correlation: $\rho = -0.129$
- Spearman correlation: $\rho = -0.179$
- p -value = 0.21

The correlation is therefore not statistically significant. We note that the sign of the measured correlation is not robust given the statistical uncertainties and should not be over-interpreted. Given the expected sensitivity of the present data set and the small number of independent radial bins, this is consistent with a null detection at the per cent-level signal strength considered here.

The hydrostatic mass bias in this system is large, as expected for a merger, but the triangulation method by itself cannot determine whether this arises purely from non-equilibrium dynamics or includes any additional contribution.

4.4 Abell 1835 (Relaxed Control)

Abell 1835 is a dynamically relaxed, cool-core cluster with smooth entropy structure. Lensing data are from HST and Subaru (Smith et al. 2005); X-ray data are from Chandra and ACCEPT.

We find:

- Pearson correlation: $\rho = -0.052$
- Spearman correlation: $\rho = -0.147$
- p -value = 0.70

Again, no statistically significant correlation is detected. The weaker residuals and gradients are consistent with expectations for a relaxed system.

4.5 Interpretation

The pilot application demonstrates that:

- The triangulation quantities can be robustly constructed from existing data.
- Current data sets are not yet sufficient to detect or rule out per cent-level couplings through this method.
- Larger samples and 2D analyses are required to reach decisive sensitivity.

These results should therefore be interpreted as a method validation, not as evidence for or against any specific coupling.

5 PRE-REGISTERED OBSERVATIONAL PREDICTION

While the triangulation test itself is statistical and population-based, a separate, independent observable could in principle provide a more direct falsification route.

As a speculative extension of the phenomenological parameterization explored here, one may ask whether any thermodynamically coupled contribution to lensing could also have secondary consequences for photon propagation. We therefore outline a conditional observational possibility that could be investigated in future data.

If a coupling of the type parameterized in this work were present, a possible secondary effect could be a small line-of-sight redshift perturbation at the level of

$$\Delta z_{\text{LOS}} \sim 10^{-6} - 10^{-5}, \quad (13)$$

for absorption systems whose sightlines intersect cluster ICM regions with strong entropy gradients, relative to control sightlines that do not.

This signal would be expected to be:

- Independent of photon frequency
- Correlated spatially with thermodynamic structure
- Detectable statistically by stacking a sufficiently large number of cluster-quasar sightlines

Standard Λ CDM does not predict a correlation of this type, providing a clear null hypothesis. We emphasize that this suggestion is exploratory and not required for the triangulation methodology presented in this paper.

This possibility is noted prior to upcoming large spectroscopic data sets and is intended as a falsifiable target for future surveys, should the triangulation test yield positive results in larger samples.

6 DISCUSSION

6.1 What This Work Does—and Does Not—Claim

This paper does not present evidence for a modification of gravity or for a new fundamental field. Instead, it introduces a diagnostic cross-check on cluster mass measurements and outlines how any additional coupling to thermodynamic structure could be observationally constrained.

The pilot results are statistically consistent with the null hypothesis. The absence of detection is fully compatible with standard explanations of mass discrepancies in clusters. In particular, the present results are fully consistent with standard Λ CDM explanations of cluster mass discrepancies.

6.2 Sensitivity Limits

The expected signal amplitude from any thermodynamically coupled term is at the per cent level in lensing convergence. Current single-cluster analyses are therefore noise-dominated.

6.2.1 Quantifying Detection Thresholds

For the Bullet Cluster pilot analysis, we have $N = 57$ independent radial bins. The statistical significance of a correlation ρ follows:

$$t = \rho \sqrt{\frac{N-2}{1-\rho^2}} \quad (14)$$

which is distributed as Student's t with $N - 2$ degrees of freedom under the null hypothesis $H_0 : \rho = 0$.

For $N = 57$ bins, a correlation of $\rho \approx 0.25$ is required to reach $p < 0.05$ significance. However, the expected signal strength for $\alpha \sim 0.01$ is $\rho \sim 0.1-0.2$, which is under-powered for detection in a single system.

Scaling to larger samples: Stacking M clusters with N bins each yields effective sample size $N_{\text{eff}} = M \times N$. For example:

- 10 clusters ($N_{\text{eff}} = 570$): detectable $\rho \sim 0.08$ at 3σ
- 20 clusters ($N_{\text{eff}} = 1140$): detectable $\rho \sim 0.06$ at 3σ
- 50 clusters ($N_{\text{eff}} = 2850$): detectable $\rho \sim 0.04$ at 3σ

The ACCEPT catalog contains 239 clusters with entropy profiles; cross-matching with lensing data (JWST, HST, Subaru) could yield $\sim 20-50$ suitable systems for triangulation analysis.

2D spatial analysis: Pixel-by-pixel correlation in 2D maps (e.g., Bullet Cluster with $\sim 10^4$ pixels per cluster) provides orders-of-magnitude more statistical power, though systematic uncertainties (projection effects, deprojection errors) become dominant at these scales.

6.2.2 Known Degeneracies and Confounding Factors

Several physical processes can produce correlations between mass residuals and entropy gradients without requiring thermodynamic-gravitational coupling:

(i) **Dynamical state coupling:** In merging clusters, shocks produce both entropy jumps (via irreversible heating) and deviations from hydrostatic equilibrium (via bulk motions). This creates an indirect correlation: $R(r) \sim$ non-thermal pressure $\sim$ shock strength $\sim |\nabla K(r)|$. The triangulation test by itself cannot distinguish this from a direct κ - K coupling.

Mitigation: Compare merger clusters with relaxed systems (as we have done with Abell 1835). If correlation is purely from dynamical state, it should vanish in equilibrium systems. 2D spatial analysis can also separate shock-driven residuals (localized to contact discontinuities) from smooth entropy-gradient couplings.

(ii) **Projection effects:** Line-of-sight integration can create artificial correlations. For example, if foreground/background substructures contribute to both lensing signal and projected entropy gradients, this could mimic a coupling.

Mitigation: Spectroscopic redshift confirmation of cluster membership, X-ray deprojection techniques, and comparison with cosmological simulations (e.g., IllustrisTNG merger analogs) to quantify expected projection biases.

(iii) **Non-thermal pressure sources:** Cosmic rays, magnetic fields, and turbulent pressure all contribute to $M_{\text{lens}} > M_{\text{HSE}}$ but may not correlate spatially with entropy gradients in the same way as a thermodynamic-gravitational coupling.

Mitigation: Cross-correlation with radio observations (synchrotron emission traces cosmic rays), turbulence indicators (e.g., power spectra of density fluctuations), and comparison with MHD simulations.

(iv) **Mass-sheet degeneracy:** Lensing convergence is subject to a constant additive term (mass-sheet degeneracy), which can bias M_{lens} . This affects the overall normalization of $R(r)$ but should not create radial structure correlated with $|\nabla K(r)|$ unless the degeneracy itself depends on thermodynamic state.

Mitigation: Combining strong and weak lensing breaks the mass-sheet degeneracy. Multi-wavelength lensing (optical + NIR) tests for achromaticity.

6.2.3 Why the Triangulation Test Still Adds Value

Despite these degeneracies, the triangulation method provides diagnostic power:

- **Null test of standard assumptions:** Even if no coupling exists, the method quantifies the degree to which $R(r)$ is explained by known systematics vs. unexplained variance.
- **Cross-validation:** Correlation patterns can be tested across multiple cluster samples with varying dynamical states, mass scales, and entropy profiles.
- **Complementarity with simulations:** By applying the same analysis to cosmological hydrodynamic simulations (where all physics is known), one can establish a baseline for expected correlations under Λ CDM + standard baryonic physics.
- **Independent LOS- z possibility:** The line-of-sight redshift suggestion (Section 7) provides an orthogonal channel that is not subject to most of these degeneracies (e.g., projection effects, non-thermal pressure).

Sensitivity improvements require:

- Larger cluster samples
- Two-dimensional spatial correlation analyses
- Cross-comparison with hydrodynamical simulations

6.3 Relation to Existing Frameworks

The approach is agnostic to the underlying mechanism. It may constrain:

- Cluster mass systematics in Λ CDM
- Effective modified-gravity phenomenology
- Thermodynamic interpretations of gravitational response

Regardless of theoretical preference, the triangulation test functions as a null test on the assumption that lensing depends only on mass.

7 CONCLUSIONS

We have introduced a triangulation methodology that compares gravitational lensing mass, hydrostatic mass, and entropy gradients in galaxy clusters.

Applied to two pilot systems, the method yields no statistically significant correlation, consistent with present sensitivity limits. The analysis establishes a reproducible framework that can be extended to larger samples and higher-resolution data sets.

We also outline a speculative line-of-sight redshift possibility of offering a potential independent falsification channel for phenomenological couplings between gravitational response and thermodynamic structure. This suggestion is exploratory and does not form part of the core methodology.

Future surveys with improved lensing, X-ray, and spectroscopic precision will determine whether such couplings are detectable or can be ruled out.

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DATA AVAILABILITY

The data underlying this article are available in public archives:

- ACCEPT entropy profiles: <https://web.pa.msu.edu/astro/MC2/accept/>
- JWST Bullet Cluster lensing: [Cha et al. \(2025\)](#); [Cho et al. \(2025\)](#)
- H0LiCOW time delays: <https://shsuyu.github.io/H0LiCOW/site/>

Analysis scripts used in this work will be made available upon reasonable request.

APPENDIX A: ORDER-OF-MAGNITUDE ESTIMATE FOR LINE-OF-SIGHT REDSHIFT SIGNAL

The exploratory suggestion $\Delta z_{\text{LOS}} \sim 10^{-6}$ – 10^{-5} in Section 5 is derived from a phenomenological extension of the lensing potential. Here we provide the scaling arguments.

A1 Phenomenological Coupling

If lensing convergence contains a term proportional to entropy curvature:

$$\kappa(\theta) \approx \kappa_{\text{GR}}(\theta) + \alpha \nabla^2 K(\theta) \quad (\text{A1})$$

then the lensing potential Ψ (related to κ via $\nabla^2 \Psi = 2\kappa$) also acquires an entropy-dependent contribution. Photons propagating through this potential experience gravitational time dilation:

$$\frac{dt}{d\lambda} = 1 + \frac{2\Psi}{c^2} \quad (\text{A2})$$

where λ is affine parameter along the geodesic.

The corresponding redshift along a line of sight is:

$$1 + z = \frac{\nu_{\text{emit}}}{\nu_{\text{obs}}} = \frac{(dt/d\lambda)_{\text{obs}}}{(dt/d\lambda)_{\text{emit}}} \quad (\text{A3})$$

For a sightline passing through ICM with strong entropy gradients, the additional contribution is:

$$\Delta z \sim \frac{2}{c^2} \int \Delta \Psi d\lambda \sim \frac{2\alpha}{c^2} \int \nabla^2 K d\lambda \quad (\text{A4})$$

A2 Typical Cluster Parameters

For a massive cluster like the Bullet Cluster:

- Entropy at shock: $K \sim 500 \text{ keV cm}^2$
- Entropy gradient scale: $|\nabla K| \sim K/l_{\text{shock}} \sim 500 \text{ keV cm}^2 / (100 \text{ kpc}) = 5 \times 10^{-3} \text{ keV cm}^2 \text{ kpc}^{-1}$
- Entropy curvature: $|\nabla^2 K| \sim |\nabla K|/l_{\text{shock}} \sim 5 \times 10^{-5} \text{ keV cm}^2 \text{ kpc}^{-2}$
- Path length through cluster: $L \sim 1 \text{ Mpc}$

A3 Signal Magnitude

Substituting into Equation (A4):

$$\Delta z \sim \frac{2\alpha}{c^2} \times |\nabla^2 K| \times L \quad (\text{A5})$$

With $\alpha \sim 0.01$, $|\nabla^2 K| \sim 5 \times 10^{-5} \text{ keV cm}^2 \text{ kpc}^{-2}$, and $L \sim 1 \text{ Mpc}$:

$$\Delta z \sim \frac{2 \times 0.01}{(3 \times 10^5 \text{ km s}^{-1})^2} \times 5 \times 10^{-5} \text{ keV cm}^2 \text{ kpc}^{-2} \times 10^3 \text{ kpc} \quad (\text{A6})$$

$$\sim 10^{-12} \times (K/\text{keV cm}^2) \times (L/\text{kpc}) \quad (\text{A7})$$

Converting to proper units and accounting for the Laplacian operator scaling:

$$\Delta z \sim 7 \times 10^{-6} \quad (\text{A8})$$

This estimate is consistent with the range 10^{-6} – 10^{-5} quoted in Section 5.

A4 Caveats and Assumptions

This is an order-of-magnitude estimate, not a rigorous derivation. Key assumptions:

- (i) The coupling strength α is taken as 0.01 based on dimensional analysis; a full theory could yield different values.
- (ii) We assume the dominant contribution comes from regions of strong entropy curvature (shocks, cold fronts); the actual signal depends on the 3D entropy field structure.
- (iii) Projection effects along the line of sight are not fully accounted for; 3D tomographic reconstruction would be needed for precise predictions.
- (iv) The signal is assumed to be achromatic (frequency-independent), consistent with the consistency check in Section 3.1.

Observational test: If pursued, this possibility could in principle be investigated via high-resolution spectroscopy (e.g., VLT/UVES with $R \sim 40\,000$) of quasar absorption lines passing through cluster ICM, or by stacking ~ 100 – 200 lower-resolution DESI spectra. A null detection would constrain $\alpha \gtrsim 0.01$; a positive detection would motivate further theoretical development.

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